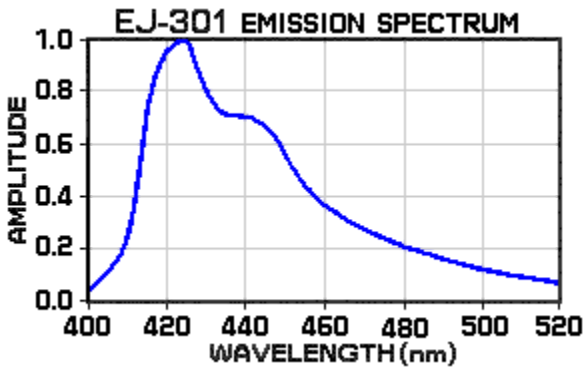
A. PbWO4 (undoped)

SY: 200 phos /MeV (Annenkov page32, table1)
SY1: 80%
SY2: 20% (science direct)
Tau1: 5ns
Tau2: 30ns (science direct)
Risetime: 10ps (Elsevier section 4.1)
RIPhotonEnergy (eV) = {1.96 *eV, 2.41 *eV, 2.50 *eV, 2.54 *eV, 2.60 *eV, 2.71 *eV}
RI: {2.24, 2.29, 2.3, 2.3, 2.31, 2.32} (Research gate PbWO4N30)

2 peaks at emission spectrum: 420nm(2.9 eV) & 480-520nm(2.5 eV)



(Image source)

But for now considering only one with peak around 420 nm.
ScinPhoEnergy eV: {2.40, 2.41, 2.42, 2.43, 2.44, 2.45, 2.46, 2.47, 2.48, 2.50, 2.51, 2.52, 2.53, 2.54, 2.55, 2.56, 2.57, 2.58, 2.60, 2.61, 2.62, 2.63, 2.64, 2.66, 2.67, 2.68, 2.69, 2.71, 2.72, 2.73, 2.75, 2.76, 2.77, 2.79, 2.80, 2.81, 2.83, 2.84, 2.86, 2.87, 2.88, 2.90, 2.91, 2.93, 2.94, 2.96, 2.97, 2.99, 3.01, 3.02}

FastScintillation Array: {0.059, 0.074, 0.065, 0.074, 0.074, 0.086, 0.083, 0.089, 0.098, 0.113, 0.110, 0.127, 0.133, 0.142, 0.151, 0.160, 0.169, 0.172, 0.187, 0.207, 0.216, 0.234, 0.243, 0.267, 0.276, 0.311, 0.341, 0.367, 0.409, 0.465, 0.507, 0.533, 0.545, 0.551, 0.551, 0.560, 0.599, 0.640, 0.726, 0.779, 0.761, 0.747, 0.687, 0.516, 0.320, 0.222, 0.136, 0.089, 0.059, 0.047}

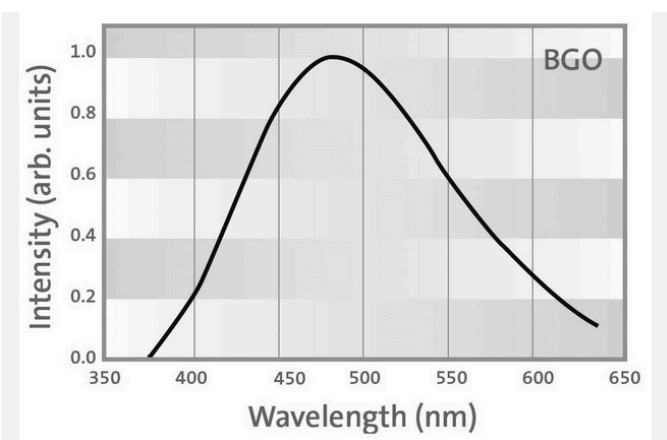
SlowScintillation Array: {0.015, 0.019, 0.016, 0.019, 0.019, 0.021, 0.021, 0.022, 0.024, 0.028, 0.027, 0.032, 0.033, 0.036, 0.038, 0.040, 0.042, 0.043, 0.047, 0.052, 0.054, 0.059, 0.061, 0.067, 0.069, 0.078, 0.085, 0.092, 0.102, 0.116, 0.127, 0.133, 0.136, 0.138, 0.138, 0.140, 0.150, 0.160, 0.181, 0.195, 0.190, 0.187, 0.172, 0.129, 0.080, 0.056, 0.034, 0.022, 0.015, 0.012}

B. BGO

SY: 8200 Phos /MeV
Birk's constant: 0.126 mm/MeV for elect
Cherenkov yield for 511 keV: 20
Rise time: 30 ps
SY1: 10%
SY2: 90%
Tau1: 60 ns
Tau2: 300 ns
Absorption length: 10.4 mm for 511 keV photon
RI: 2.39 (310 nm), 2.2 (420 nm) (Research gate table)
Peak at: wavelength 480 nm (ref)
RI at peak: 2.15 (refractiveindex.info)

ScintPhotonEnergy (eV): {1.95, 2.01, 2.08, 2.24, 2.51, 2.59, 2.69, 3.09, 3.40, 3.66}
Fast Scint array: {0.0213, 0.038, 0.0517, 0.0727, 0.0973, 0.0913, 0.078, 0.05, 0.0333, 0.02}
Slow Scint Array: {0.19, 0.34, 0.47, 0.65, 0.88, 0.82, 0.70, 0.45, 0.30, 0.18}

Ref:
Emission spectrum



Source

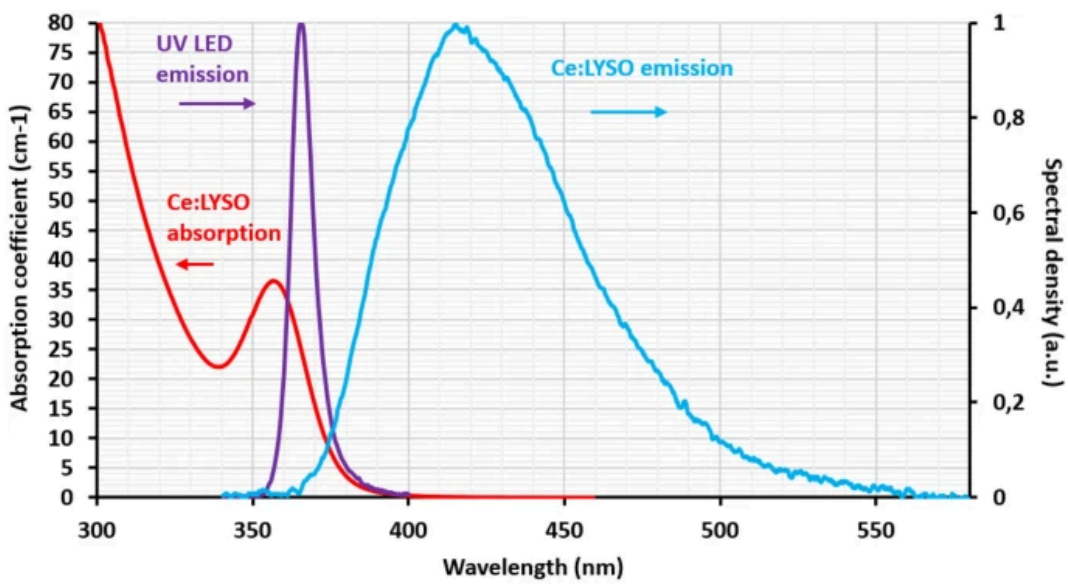
Then extracted the datapoint using some tools. (data points)

C. LYSO(Ce)

SY: 30000 Phos /MeV
Density: 7.1 g/cc
Rise time: 70 ps ref
SY1: 100 % Luxium solutions
SY2: Nil
Tau1: 40 ns
Tau2: Nil
RI:
Absorption length:
Fast Scint array:
Slow Scint Array:

Peak emission at: 410 nm (crylink)

Emission spectrum:



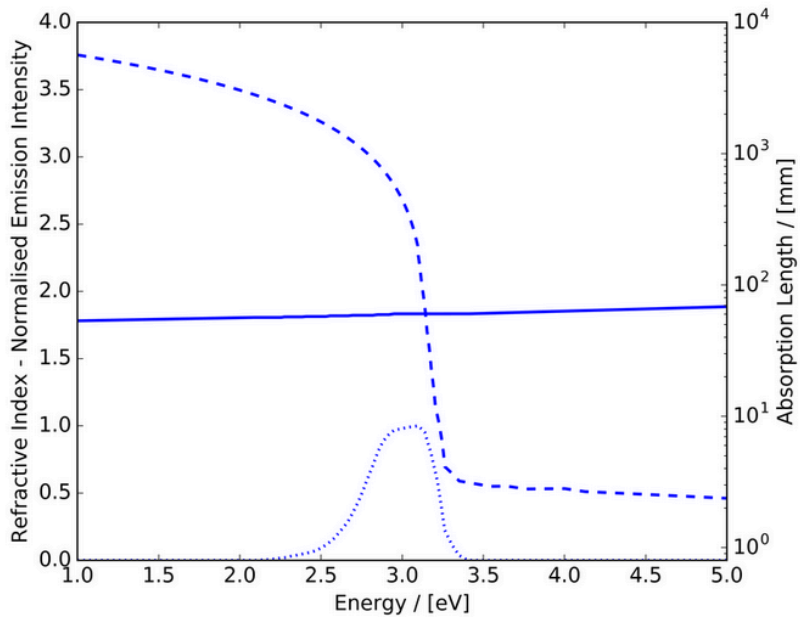
Taking the point from the **cyan** line.
source: Nature

SciPhoEnergy (eV):
{2.189,2.199,2.212,2.222,2.234,2.254,2.268,2.285,2.293,2.316,2.330,2.350,2.366,2.373
,2.380,2.396,2.408,2.411,2.428,2.444,2.450,2.458,2.471,2.493,2.508,2.524,2.534,2.538
,2.546,2.573,2.588,2.593,2.603,2.619,2.634,2.650,2.657,2.668,2.687,2.714,2.753,2.784
,2.815,2.824,2.839,2.880,2.914,2.924,2.940,2.940,2.954,2.962,2.970,2.984,2.989,3.001

,3.018,3.028,3.037,3.090,3.111,3.144,3.165,3.195,3.229,3.247,3.263,3.285,3.314,3.354
,3.385,3.400,3.418,3.435,3.513,3.559,3.635}

Rel Intensity (a.u.)
{0.016,0.004,0.023,0.002,0.019,0.026,0.021,0.019,0.035,0.033,0.035,0.047,0.055,0.039
,0.052,0.055,0.070,0.064,0.082,0.084,0.103,0.100,0.112,0.127,0.156,0.167,0.187,0.204
,0.191,0.244,0.267,0.284,0.289,0.323,0.356,0.365,0.394,0.411,0.444,0.504,0.602,0.698
,0.760,0.791,0.819,0.887,0.931,0.941,0.966,0.951,0.960,0.990,0.980,0.990,0.996,0.980
,0.969,0.934,0.929,0.803,0.754,0.658,0.594,0.501,0.372,0.309,0.250,0.185,0.106,0.050
,0.036,0.008,0.017,0.006,0.016,0.006,0.006}

<u>For RI:</u>



(taking the points from blue solid line in the image)
source: [Reasearch gate]

RI PhoEnergy (eV):
{0.996,1.177,1.425,1.673,1.968,2.304,2.643,2.841,2.949,3.044,3.130,3.215,3.328,3.432
,3.680,4.036,4.383,4.789}

RI:
{1.784,1.784,1.789,1.797,1.805,1.808,1.816,1.824,1.833,1.835,1.833,1.838,1.835,1.835
,1.841,1.852,1.868,1.882}

==========

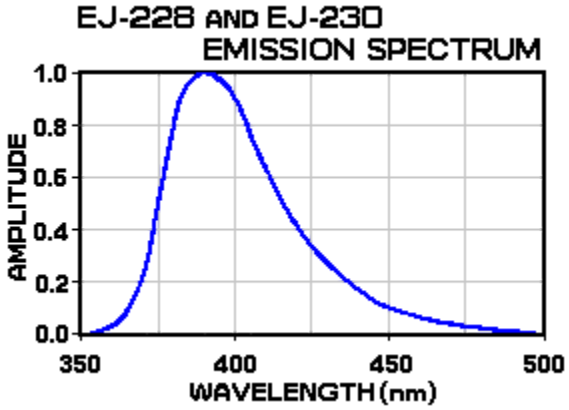
D. EJ-230 plastic scintillator (used in JPET)

SY: 9700 phos/ MeV e-
Rise time: 0.5 ns
No significant slow component
SY1: 1
Tau1:1.5 ns
RI: 1.58 (defining this for all photon energies)
Absorption length:

Peak at: 391 nm (Eligen Technology)

Required things for defining PolyVinyl toluene for EJ230:
Chemical formula: C9H10
Density 1.032 g/cc
Component fractions: C=0.915, H=0.085 (IOP paper on PVT Scintillator)

Emission spectrum



source: Eligen

Backup for PbWO4
Using Nb doped crystal
SY: 200 phos /MeV (Annenkov page32, table1)
SY1: 55, SY2: 45 (lecoq section 3.1)
Tau1: 3ns, tau2: 14ns (lecoq section 3.1)

==========

GAGG(Ce) ($Gd_3Al_2Ga_3O_{12}$: Gadolinium Aluminium Gallium Garnet)

Mainly Used Reference: Science Direct
There are two types Single crystal GAGG:Ce and ceramic GAGG:Ce. Here I am using single crystal one.

Density: 6.63 g/cm³ (ref)

RI: points are extracted from the image

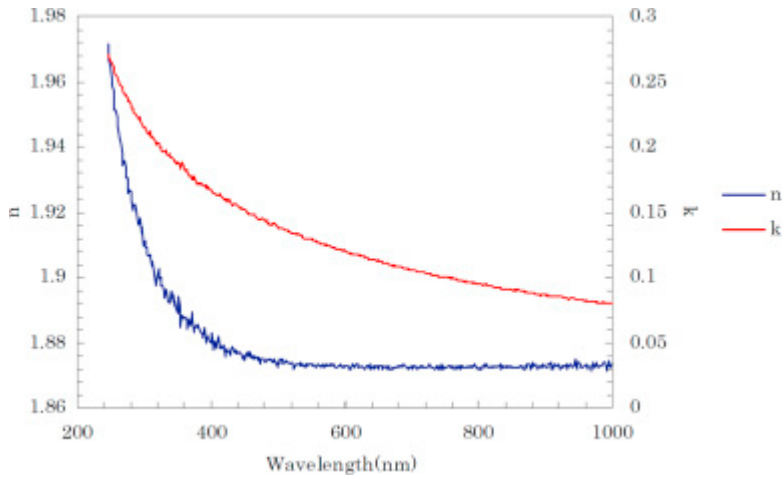


image source

RIPhotonEnergy (eV): {1.56, 2.19, 2.38, 3.47, 4.24, 4.65}
RI: {1.872, 1.873, 1.870, 1.887, 1.914, 1.940}

Rise time: 8ns (see table 3 here)
SY: 28244 pho/MeV (depends on size of crystal this one is 5mm*5mm*5mm)

Crystal has one strong peak at 520 nm (many PMTs can't process this wavelength) and a weak peak at 380 nm. Both the peaks slow scintillation and fast scintillation components are different (can found in the reference given above.)
Since I am not aware of providing multiple fast and slow components, I am using only strong peak and decay constants corresponding to that.

tau1: 50.1 ns
SY1: 39.2%

tau2: 321.5 ns
SY2: 60.8%

Note: *the value of SY1 and SY2 are different given in the table3 of the paper. Right now I am taking the values from main reference provided at the beginning.*

Emission spectrum:

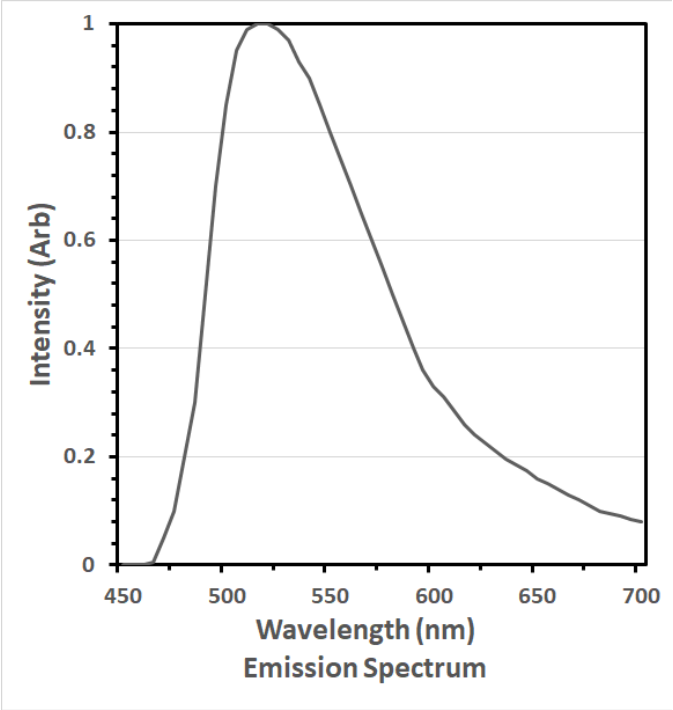


image source

ScintPhoton Energy (eV): {1.771, 1.791, 1.816, 1.834, 1.857, 1.878, 1.901, 1.916, 1.933, 1.951, 1.970, 1.988, 2.010, 2.025, 2.040, 2.056, 2.074, 2.091, 2.106, 2.122, 2.133, 2.143, 2.155, 2.164, 2.174, 2.181, 2.190, 2.196, 2.207, 2.217, 2.230, 2.243, 2.254, 2.264, 2.274, 2.286, 2.296, 2.310, 2.319, 2.329, 2.342, 2.354, 2.372, 2.389, 2.405, 2.421, 2.431, 2.444, 2.450, 2.453, 2.460, 2.467, 2.471, 2.475, 2.479, 2.485, 2.491, 2.496, 2.499, 2.506, 2.506, 2.510, 2.515, 2.519, 2.522, 2.526, 2.531, 2.536, 2.539, 2.543, 2.548, 2.557, 2.565, 2.571, 2.580, 2.589, 2.598, 2.613, 2.631, 2.645, 2.665}

FastScintArray: {0.032, 0.036, 0.039, 0.045, 0.051, 0.057, 0.063, 0.069, 0.073, 0.078, 0.086, 0.093, 0.104, 0.112, 0.122, 0.128, 0.140, 0.157, 0.174, 0.191, 0.201, 0.214, 0.225, 0.236, 0.244, 0.253, 0.261, 0.268, 0.278, 0.289, 0.301, 0.313, 0.324, 0.334, 0.343, 0.353, 0.360, 0.366, 0.374, 0.381, 0.385, 0.389, 0.392, 0.392, 0.391, 0.388, 0.381, 0.373, 0.364, 0.355, 0.345, 0.335, 0.324, 0.314, 0.306, 0.292, 0.279, 0.266, 0.254, 0.231, 0.236, 0.218, 0.204, 0.193, 0.183, 0.170, 0.157, 0.145, 0.133, 0.121, 0.110, 0.099, 0.087, 0.078, 0.065, 0.051, 0.039, 0.028, 0.017, 0.008, 0.002}

SlowScintArray: {0.050, 0.055, 0.061, 0.069, 0.079, 0.088, 0.097, 0.107, 0.114, 0.122, 0.134, 0.144, 0.161, 0.174, 0.189, 0.199, 0.217, 0.244, 0.270, 0.296, 0.312, 0.332, 0.348, 0.365, 0.379, 0.392, 0.405, 0.416, 0.432, 0.448, 0.466, 0.486, 0.502, 0.517, 0.533, 0.548, 0.558, 0.568, 0.579, 0.591, 0.598, 0.603, 0.608, 0.608, 0.606, 0.602, 0.591, 0.578, 0.565, 0.551, 0.535, 0.519, 0.503, 0.486, 0.474, 0.453, 0.433, 0.413, 0.394, 0.358, 0.365, 0.339, 0.317, 0.300, 0.284, 0.264, 0.244, 0.224, 0.207, 0.188, 0.171, 0.153, 0.136, 0.121, 0.100, 0.079, 0.061, 0.044, 0.027, 0.013, 0.003}

Optional references:
palit et al
avatech-uk
https://www.sciencedirect.com/science/article/pii/S0925346718301691

==========

LaBr$_3$:Ce

Choosing the 0.5% doping because there afterglow not observed (Ref: Section 2)
Density: 5.29 g/cm3 (Ref Sec. 2)
The crystal is Hygroscopic

Refractive index:

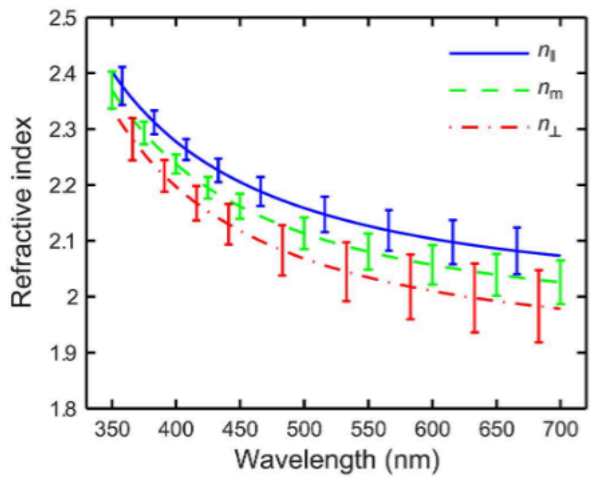


Fig. 4. The experimentally determined refractive index of $LaBr_3$:5%Ce^{3+}. The error bars correspond to the standard deviations, which were obtained as described in Section III-A. The component $n_\perp$ is depicted in red (dash-dotted curve), $n_\parallel$ in blue (solid curve), and their mean n_m in green (dashed curve).

image source (taking the mean n)

RIPhoEnergy (eV): {1.772, 1.908, 2.066, 2.255, 2.481, 2.757, 2.919, 3.102, 3.305, 3.545}

RI: {2.025, 2.039, 2.057, 2.080, 2.113, 2.162, 2.195, 2.237, 2.291, 2.371}
(The RI at max emission described in the paper is higher than that of many references)
Rise Time = 5.7 ns (ref: section 2.2)
SY: 61000 pho/MeV (Ref: table1)
SY1: 90%
Tau1: 30ns
SY2: 10%
Tau2: 300ns (Ref: section2)

Emission Spectrum:

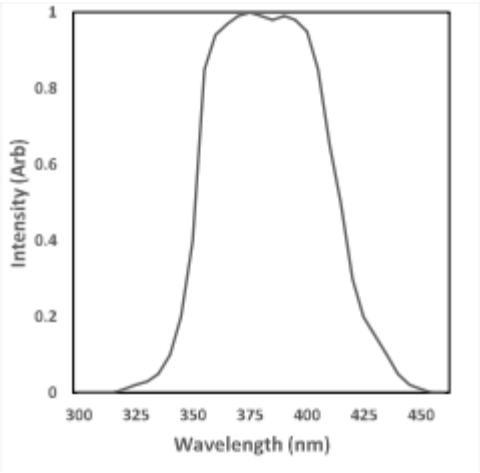


image source

Scint Pho Energy: {2.813, 2.863, 2.891, 2.919, 2.948, 2.982, 2.997, 3.010, 3.020, 3.028, 3.038, 3.044, 3.049, 3.060, 3.067, 3.077, 3.085, 3.093, 3.103, 3.113, 3.121, 3.131, 3.145, 3.160, 3.188, 3.236, 3.272, 3.302, 3.350, 3.400, 3.436, 3.483, 3.504, 3.520, 3.529, 3.531, 3.536, 3.544, 3.553, 3.559, 3.563, 3.574, 3.587, 3.596, 3.613, 3.635, 3.663, 3.713, 3.829, 3.906}

FastScintArray: {0.002, 0.023, 0.046, 0.090, 0.131, 0.183, 0.212, 0.246, 0.282, 0.327, 0.376, 0.409, 0.448, 0.481, 0.508, 0.546, 0.582, 0.628, 0.665, 0.731, 0.763, 0.790, 0.819, 0.853, 0.878, 0.890, 0.879, 0.886, 0.898, 0.885, 0.871, 0.840, 0.798, 0.763, 0.731, 0.699, 0.649, 0.591, 0.531, 0.466, 0.412, 0.333, 0.298, 0.258, 0.198, 0.145, 0.094, 0.045, 0.018, 0.002}

SlowScintArray: {0.000, 0.003, 0.005, 0.010, 0.015, 0.020, 0.024, 0.027, 0.031, 0.036, 0.042, 0.045, 0.050, 0.053, 0.056, 0.061, 0.065, 0.070, 0.074, 0.081, 0.085, 0.088, 0.091, 0.095, 0.098, 0.099, 0.098, 0.098, 0.100, 0.098, 0.097, 0.093, 0.089, 0.085, 0.081, 0.078, 0.072, 0.066, 0.059, 0.052, 0.046, 0.037, 0.033, 0.029, 0.022, 0.016, 0.010, 0.005, 0.002, 0.000}

Optional Ref: AdvaTech
****

Emission spectrum for BGO to correct the zittring

Photon Energies (eV)

{1.952,1.967,1.995,2.020,2.035,2.048,2.069,2.101,2.122,2.137,2.151,2.180,2.202,2.226,2.249,2.274,2.302,2.327,2.352,2.376,2.403,2.436,2.463,2.493,2.534,2.567,2.601,2.628,2.648,2.665,2.686,2.709,2.726,2.744,2.767,2.787,2.807,2.826,2.841,2.855,2.874,2.893,2.914,2.936,2.953,2.977,3.000,3.025,3.052,3.093,3.121,3.155,3.195,3.249,3.302}

Fast scintillation array:

{0.011,0.012,0.016,0.019,0.021,0.023,0.027,0.031,0.035,0.038,0.040,0.045,0.049,0.054,0.058,0.063,0.069,0.074,0.079,0.083,0.087,0.092,0.095,0.098,0.100,0.101,0.100,0.098,0.097,0.095,0.093,0.090,0.088,0.085,0.081,0.078,0.074,0.070,0.067,0.063,0.060,0.056,0.052,0.048,0.044,0.040,0.035,0.031,0.026,0.020,0.017,0.014,0.010,0.005,0.000}

Slow Scintillation array:

{0.098,0.112,0.142,0.174,0.193,0.211,0.242,0.283,0.317,0.338,0.361,0.406,0.445,0.485,0.524,0.570,0.622,0.669,0.711,0.747,0.787,0.826,0.853,0.880,0.898,0.905,0.898,0.883,0.870,0.854,0.835,0.810,0.788,0.764,0.732,0.702,0.664,0.631,0.601,0.571,0.538,0.503,0.467,0.431,0.397,0.356,0.317,0.276,0.234,0.184,0.153,0.124,0.087,0.044,0.001}